

DECIMOMANNU, Italy

WMO: 165460

Lat: 39.3459N

Lon: 8.9704E

Elev: 100

StdP: 14.64

Time Zone: 1.00 (EUC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.6	35.7	26.2	20.2	52.1	29.5	23.7	48.3	30.1	56.5	26.3	55.7	3.1	330	0.467

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.3	96.9	71.6	94.6	71.5	91.6	70.8	79.6	87.4	77.9	85.6	76.2	83.9	12.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	142.8	83.7	75.5	134.0	82.1	73.8	126.3	80.5	43.2	87.2	41.4	85.5	39.7	83.9	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.0	22.1	19.8	DB	29.5	103.6	2.2	3.6	28.0	106.2	26.7	108.2	25.5	110.2	23.9	112.8
			WB	28.8	82.7	2.1	2.7	27.3	84.7	26.1	86.2	24.9	87.7	23.4	89.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.7	49.6	49.9	54.3	59.2	65.7	74.3	79.9	79.8	73.4	67.2	58.1	51.9	(6)
	DBStd	11.80	4.57	4.70	4.81	4.15	4.76	5.15	4.01	3.99	4.49	4.92	5.14	4.88	(7)
	HDD50	176	62	55	18	0	0	0	0	0	0	0	5	36	(8)
	HDD65	2110	476	424	333	177	48	1	0	0	1	33	211	406	(9)
	CDD50	5177	51	51	150	278	486	730	928	923	701	535	249	95	(10)
	CDD65	1635	0	0	0	4	69	281	463	458	252	103	5	0	(11)
	CDH74	16246	0	0	9	63	683	2896	5176	4909	1936	562	12	0	(12)
	CDH80	7188	0	0	3	2	171	1236	2621	2445	625	85	0	0	(13)
(14) Wind	WSAvg	7.9	6.3	6.9	7.8	7.9	9.3	9.7	10.6	9.1	7.6	6.9	6.1	6.1	(14)
(15) Precipitation	PrecAvg	20.9	2.0	2.1	1.9	2.1	1.2	0.6	0.3	0.4	1.5	2.2	3.7	2.9	(15)
	PrecMax	29.7	5.7	5.6	4.5	5.3	3.1	2.8	0.8	1.7	4.2	5.7	12.2	6.1	(16)
	PrecMin	12.2	0.4	0.4	0.2	0.4	0.2	0.1	0.1	0.0	0.1	0.3	0.6	0.8	(17)
	PrecStd	4.5	1.3	1.4	1.0	1.3	0.7	0.7	0.2	0.4	1.1	1.4	2.5	1.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.1	66.6	73.5	79.0	88.2	97.1	100.5	101.7	93.1	86.0	75.4	68.4	(19)
		MCWB	60.0	56.9	60.2	61.8	63.7	70.7	71.3	72.2	71.2	70.9	65.3	61.5	(20)
	2%	DB	62.7	64.1	69.7	75.5	84.5	93.3	96.8	97.2	89.3	82.3	72.0	64.8	(21)
		MCWB	57.4	55.8	57.9	59.6	63.8	69.3	72.2	71.4	71.6	68.9	64.1	59.2	(22)
	5%	DB	60.6	61.2	66.6	73.0	80.9	90.0	94.6	94.7	86.2	79.3	70.0	62.9	(23)
		MCWB	55.6	54.3	56.7	58.8	62.7	68.5	71.7	72.0	71.3	68.6	63.1	58.3	(24)
	10%	DB	58.7	59.2	64.4	69.7	77.4	87.5	91.7	91.4	83.9	77.0	67.9	60.9	(25)
		MCWB	54.1	53.2	56.1	59.0	61.7	67.8	71.1	71.6	70.1	67.6	61.8	56.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.3	60.4	63.3	67.5	71.9	78.8	82.3	82.6	79.1	75.9	68.8	62.9	(27)
		MCDB	64.4	64.5	68.7	73.9	78.9	87.8	89.9	89.7	85.4	80.8	72.9	66.6	(28)
	2%	WB	58.6	57.5	60.5	64.3	68.7	75.9	79.5	79.9	76.8	73.1	66.6	60.7	(29)
		MCDB	61.6	61.5	66.2	70.6	76.4	84.5	87.4	87.8	83.0	78.2	70.8	64.0	(30)
	5%	WB	56.4	55.8	58.9	62.3	66.6	73.8	77.5	78.1	75.0	71.3	64.4	59.0	(31)
		MCDB	59.6	59.8	64.4	68.3	74.6	82.5	85.4	86.0	81.2	76.3	68.6	62.3	(32)
	10%	WB	54.5	54.2	57.1	60.5	64.9	71.9	75.8	76.4	73.1	69.4	62.4	57.3	(33)
		MCDB	57.8	58.4	62.1	66.2	73.2	80.9	83.9	84.4	79.8	74.7	66.6	60.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	16.7	17.7	18.6	19.9	21.5	23.1	23.3	23.3	20.4	19.2	16.8	16.3	(35)
		MCDBR	18.8	20.5	22.7	25.1	27.1	27.0	27.0	27.3	23.1	21.5	19.3	18.5	(36)
	5% WB	MCWBR	14.7	14.3	13.7	12.5	11.0	10.4	11.4	12.0	11.9	12.7	13.4	14.5	(37)
		MCWBR	17.4	19.2	19.2	20.4	22.6	23.6	24.2	23.9	20.4	18.8	17.5	17.3	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.346	0.366	0.406	0.445	0.460	0.435	0.449	0.438	0.435	0.410	0.382	0.353	(40)
		taud	2.426	2.353	2.237	2.142	2.138	2.258	2.222	2.260	2.261	2.302	2.333	2.392	(41)
	Ebn at Noon	Ebn at Noon	259	269	268	264	262	268	264	263	256	249	240	245	(42)
		Edh at Noon	27	33	40	47	49	43	44	42	39	34	29	26	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	655	937	1344	1728	2073	2390	2418	2121	1588	1138	726	591	(44)
	RadStd	RadStd	60	84	138	121	148	81	77	101	58	81	52	52	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	+0.52	N/A	N/A	N/A	-0.90	-1.10	N/A	N/A	N/A	N/A

Nomenclature: See separate page